

Theory Week 13

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Orga

Mechanics 3 CAB G59
WhatsApp group



- Find all doc's on my website (link in WhatsApp bio)
- remember to do the STICK
- Study Center
→ Wednesday 18:00-20:00 (ETC &)

13	09/12/24	free/damped/forced multi-DOF vibrations	4.4.3
	11/12/24*	deformable bodies, effective spring models	5.1
exercise 13: multi-DOF vibrations, effective spring models			

Damped & forced vibrations

$$\underline{\underline{M}} \ddot{\underline{x}} + \underline{\underline{C}} \dot{\underline{x}} + \underline{\underline{K}} \underline{x} = \underline{F}^{\text{ext}}(t)$$

like for single DOF, the soln is sought of form: $\underline{x}(t) = \underline{x}_{\text{part}}(t) + \underline{x}_{\text{hom}}(t)$

Lo to find $\underline{x}_{\text{hom}}(t)$: EOM of form $\underline{\underline{M}} \ddot{\underline{x}} + \underline{\underline{C}} \dot{\underline{x}} + \underline{\underline{K}} \underline{x} = 0$ (free vibration)

$$\underline{\underline{M}} = \frac{\partial^2 T}{\partial \dot{q}_i \partial \dot{q}_j}(q_0, 0), \quad \underline{\underline{C}} = -\frac{\partial Q^{\text{nc}}}{\partial \dot{q}}(q_0, 0), \quad \underline{\underline{K}} = \frac{\partial^2 V}{\partial q \partial q}(q_0) - \frac{\partial Q^{\text{nc}}}{\partial q}(q_0, 0)$$

eg. 2 DOF vibration with q_1 & q_2 :

$$\underline{\underline{M}} = \begin{pmatrix} \frac{\partial^2 T}{\partial \dot{q}_1^2} & \frac{\partial^2 T}{\partial \dot{q}_1 \partial \dot{q}_2} \\ \frac{\partial^2 T}{\partial \dot{q}_2 \partial \dot{q}_1} & \frac{\partial^2 T}{\partial \dot{q}_2^2} \end{pmatrix}, \quad \underline{\underline{K}} = \begin{pmatrix} \frac{\partial^2 V}{\partial q_1^2} & \frac{\partial^2 V}{\partial q_1 \partial q_2} \\ \frac{\partial^2 V}{\partial q_2 \partial q_1} & \frac{\partial^2 V}{\partial q_2^2} \end{pmatrix} - \begin{pmatrix} \frac{\partial Q^{\text{nc}}}{\partial q_1} & \frac{\partial Q^{\text{nc}}}{\partial q_2} \\ \frac{\partial Q^{\text{nc}}}{\partial q_1} & \frac{\partial Q^{\text{nc}}}{\partial q_2} \end{pmatrix}, \quad \underline{\underline{C}} = \begin{pmatrix} -\frac{\partial Q^{\text{nc}}}{\partial \dot{q}_1} & -\frac{\partial Q^{\text{nc}}}{\partial \dot{q}_2} \\ -\frac{\partial Q^{\text{nc}}}{\partial \dot{q}_1} & -\frac{\partial Q^{\text{nc}}}{\partial \dot{q}_2} \end{pmatrix}$$

$\underline{\underline{C}}$: structure depends on non-cons. forces

□ vel. proportional damping $\left(\begin{array}{l} \text{dashpot between particles /} \\ \text{linear viscous drag in fluid} \end{array} \right)$

Lo Rayleigh damping: $\underline{\underline{C}} = \alpha \underline{\underline{M}} + \beta \underline{\underline{K}}, \alpha, \beta \geq 0$

This is a first-order approximation; constants α and β can be tuned to damp low- and high-frequency vibrations, respectively. For example, if particles of equal mass m are submerged in a fluid to experience linear viscous drag, then we consider drag forces $Q_i^{\text{nc}} = F_{\text{drag},i} = -d\dot{x}_i$ with some damping coefficient $d > 0$, so that $\underline{\underline{C}} = d\underline{\underline{I}} = \frac{d}{m}\underline{\underline{M}} = \alpha\underline{\underline{M}}$, since for identical particles $\underline{\underline{M}} = m\underline{\underline{I}}$. Alternatively, if elastically coupled particles are interconnected by dashpots (one dashpot in parallel to each spring connection, and the dashpot viscosity between particles i and j is denoted by $d_{ij} = \beta k_{ij}$), then we have $\underline{\underline{C}} = \beta\underline{\underline{K}}$. (4.145) presents a linear combination of these two cases.

General solns for $n-l \leq n$ rigid body modes & l eigenfreq. $\neq 0$

(h.d.c. free & undamped)

eigenfrequencies ω_j and eigenmodes $\hat{x}_j$ are obtained from

$$(I) \det(-\omega_j^2 M + K) = 0 \quad \text{and} \quad (II) (-\omega_j^2 M + K) \hat{x}_j = 0$$

general solution of free vibrations for multiple-DOF systems:

$$x(t) = \sum_{j=1}^n x_j(t) = \sum_{j=1}^l c_j \cos(\omega_j t + \varphi_j) \hat{x}_j + \sum_{j=l+1}^n (a_j + b_j t) \hat{x}_j$$

$$a_j, b_j, c_j \in \mathbb{R}$$

→ see example 4.9 / 4.10 in script
simple
more difficult

use I: $\omega_j = \pm \sqrt{\omega_j^2}$ (only positive soln of relevance)

plug in found ω_j from (I) & solve lin. equation (II)

↳ always come in pair
(non-zero eigenfreq.)

↳ zero eigenfreq. $\Rightarrow$ rigid body motion

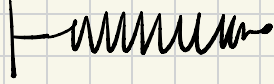
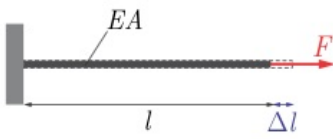
(always as many as indep. DOFs $\rightarrow$ e.g. 2D: 2 transl. & 1 rot. = 3)

Each of the ω_j is an eigenfrequency, which has its corresponding eigenvector $\hat{x}_j$ as its mode shape or eigenmode of the system

5. Dynamics of deformable bodies

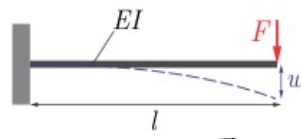
5.1 Massless deformable bodies

Bar:



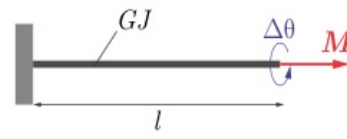
$$k_{\text{eff}} = \frac{F}{\Delta l} = \boxed{\frac{EA}{l}}$$

Beam:



$$k_{\text{eff}} = \frac{F}{w} = \boxed{\frac{3EI}{l^3}}$$

Torsion rod:



$$k_{\text{eff}} = \frac{M}{\Delta \theta} = \boxed{\frac{GJ}{l}}$$

↳ e.g. 5.2 (p. 204)

approx.
as spring

Side note: springs

